Geodesic Curvature & Euler Characteristic

The Polyakov action is given by,

$$S_X = \frac{1}{4\pi\alpha'} \int_M d^2\sigma \sqrt{g}\, g^{ab} (\partial_a X^\mu)(\partial_b X_\mu)$$

where $M \rightarrow$ Euclidean manifold, g_{ab} is the world sheet metric, $X^\mu \rightarrow$ massless scalar field.

This action has diffeomorphism and Weyl symmetry with respect to the dynamical world sheet metric discussed before which are the gauge symmetries.

Let us generalize the Polyakov action by introducing new terms to it but requiring all the symmetries to still be maintained.

One such allowed term is given by the Euler characteristic of the world sheet.

For a World sheet with or without any end point, it is given by,

$$\chi = \frac{1}{4\pi}\int_{M} d^2\sigma \sqrt{g}\, R + \frac{1}{2\pi}\int_{\partial M} ds\, k$$

M ↳ manifold $\qquad$ ∂M ↳ Boundary

where 'R' is the Ricci Scalar of the World Sheet metric, $k = t^a n_b \nabla_a t^b$ is the geodesic curvature of the boundary & $ds = \sqrt{g_{11}}\, d\sigma^1$ for the σ^2 boundary & $ds = \sqrt{g_{22}}\, d\sigma^2$ is the euclidean time along the σ^1 boundary. t^a is the tangent unit vector & n^a is a unit outward normal vector orthogonal to t^a at the boundary.

This has all the Symmetries present in the Polyakov action. Diffeomorphism Symmetry is present as all factors are world Sheet Scalars. It also has Weyl Symmetry, prooved in Chapter 1.2 of polchinski notes.

The Polyakov Path Integral

The total action that we consider is,

$$S = S_X + \lambda \chi$$

$$S_X = \frac{1}{4\pi\alpha'} \int_M d^2\sigma \sqrt{g}\, g^{ab} (\partial_a X^\mu)(\partial_b X_\mu)$$

& $$\chi = \frac{1}{4\pi} \int_M d^2\sigma \sqrt{g}\, R + \frac{1}{2\pi} \int_{\partial M} ds\, k$$

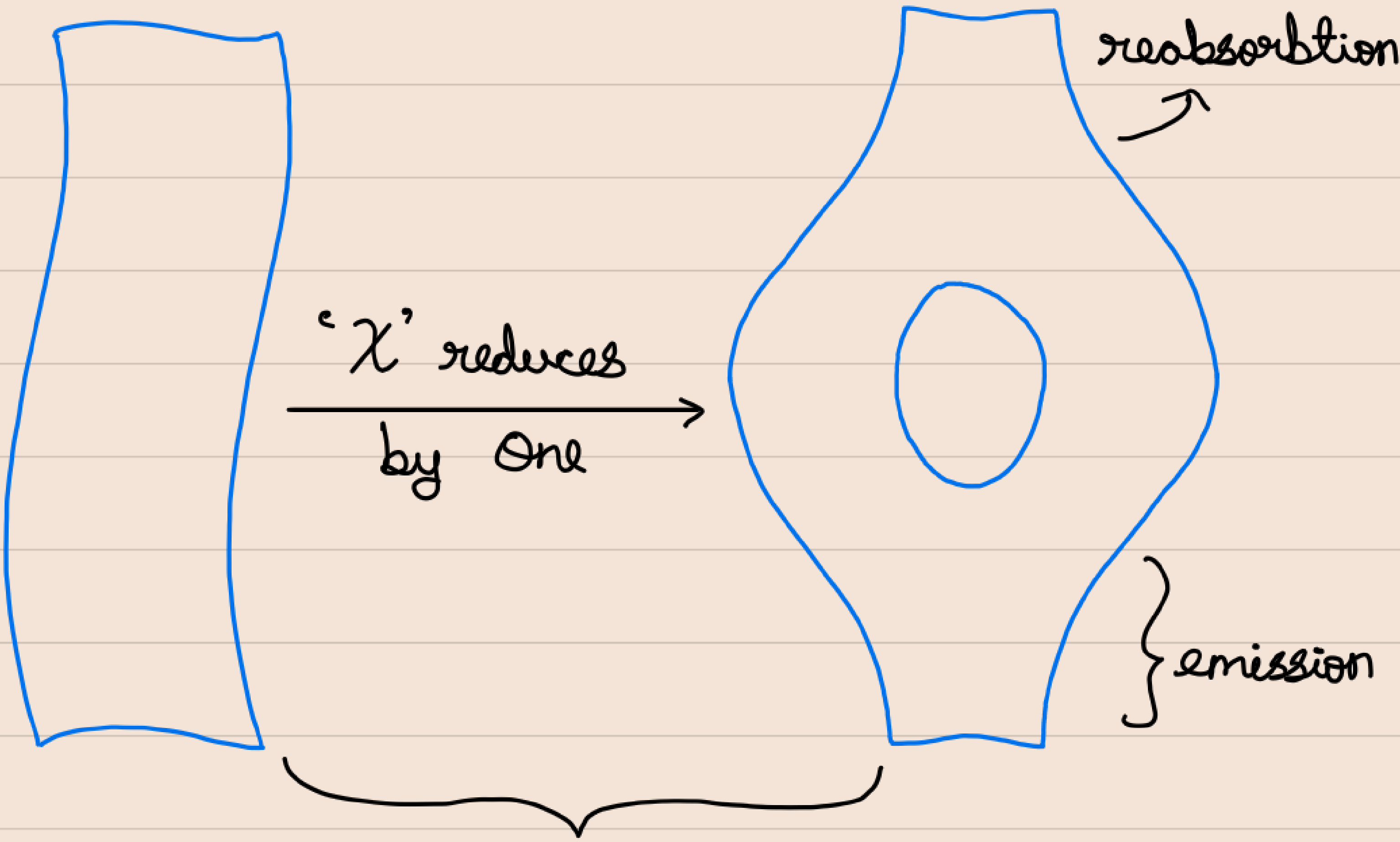
Then, the Polyakov Path integral is given by,

$$\int [\mathcal{D}X^\mu \mathcal{D}g_{ab}]\, e^{-S} = \int [\mathcal{D}X^\mu \mathcal{D}g_{ab}]\, e^{-S_X}\, e^{-\lambda\chi}$$

'χ' is a topological invariant & does not depend on the metric and X^μ fields. So, objects related by smooth deformations i.e., with same topology can have same χ, but different g_{ab}. 'χ' gives the Euler number of the world sheet.

So, $e^{-\lambda\chi}$ factor affects only the relative weighting of different topologies in the sum over world sheets.

For Open Strings

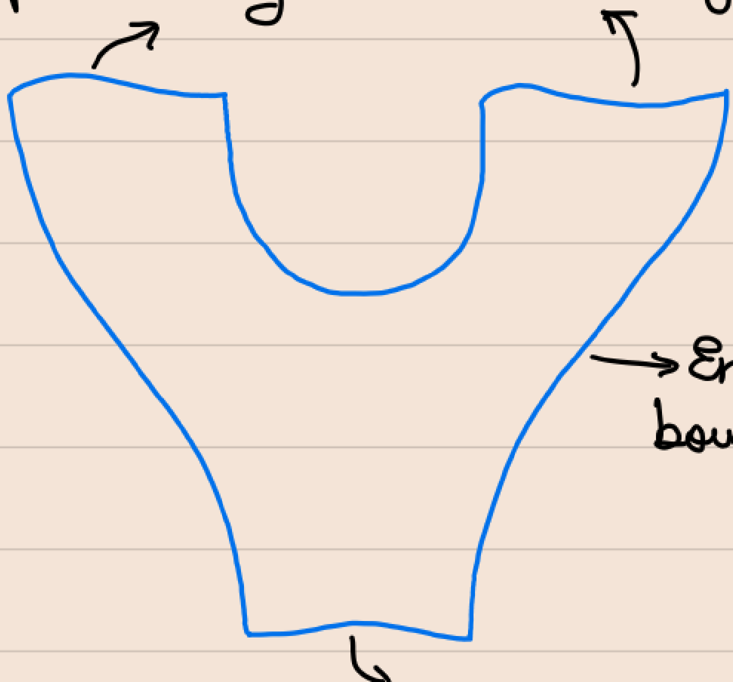


Path integral is weighted by an extra e^{λ} factor. Adding an extra strip accounts to emitting & then reabsorbing a virtual open string.

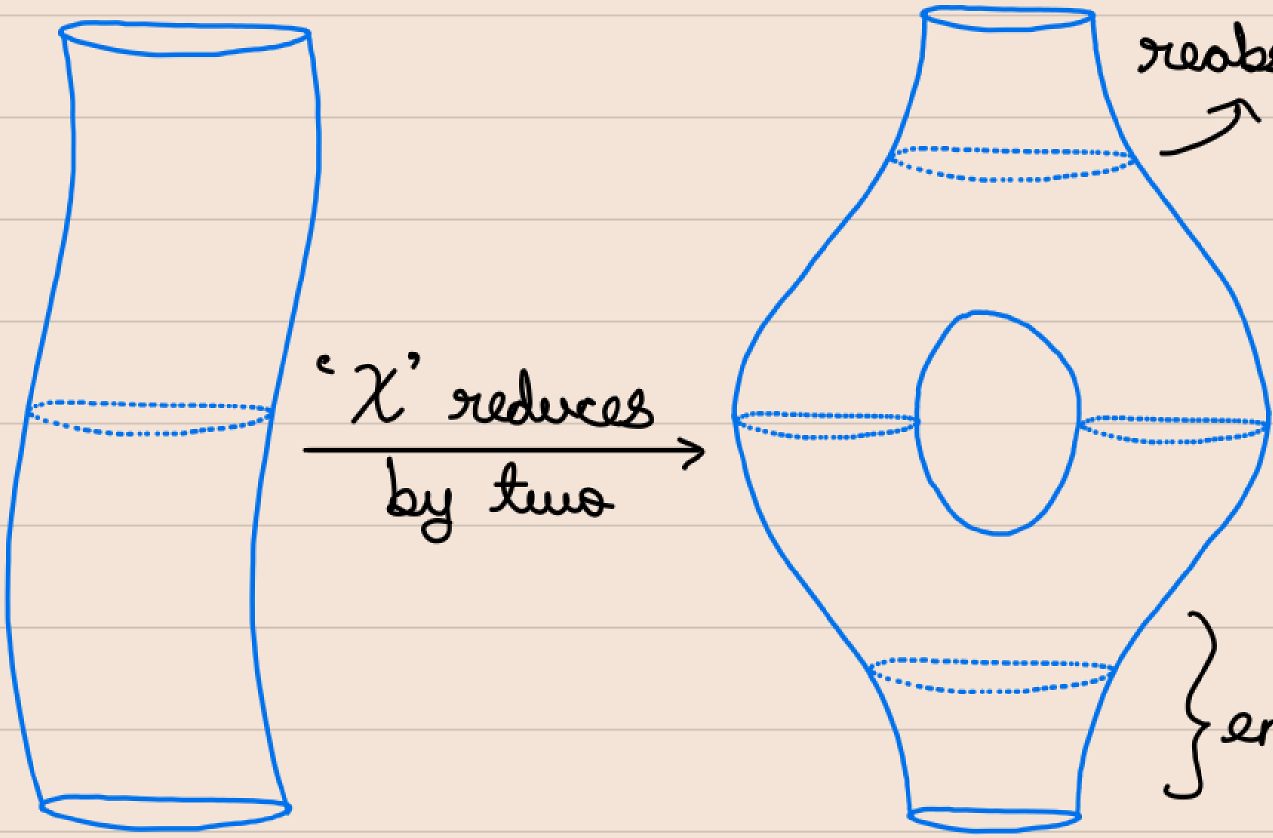
Thus amplitude of emission of open-string from any process is proportional to $e^{\lambda/2}$.

Emission of open string

$g_o \sim e^{\lambda/2}$.

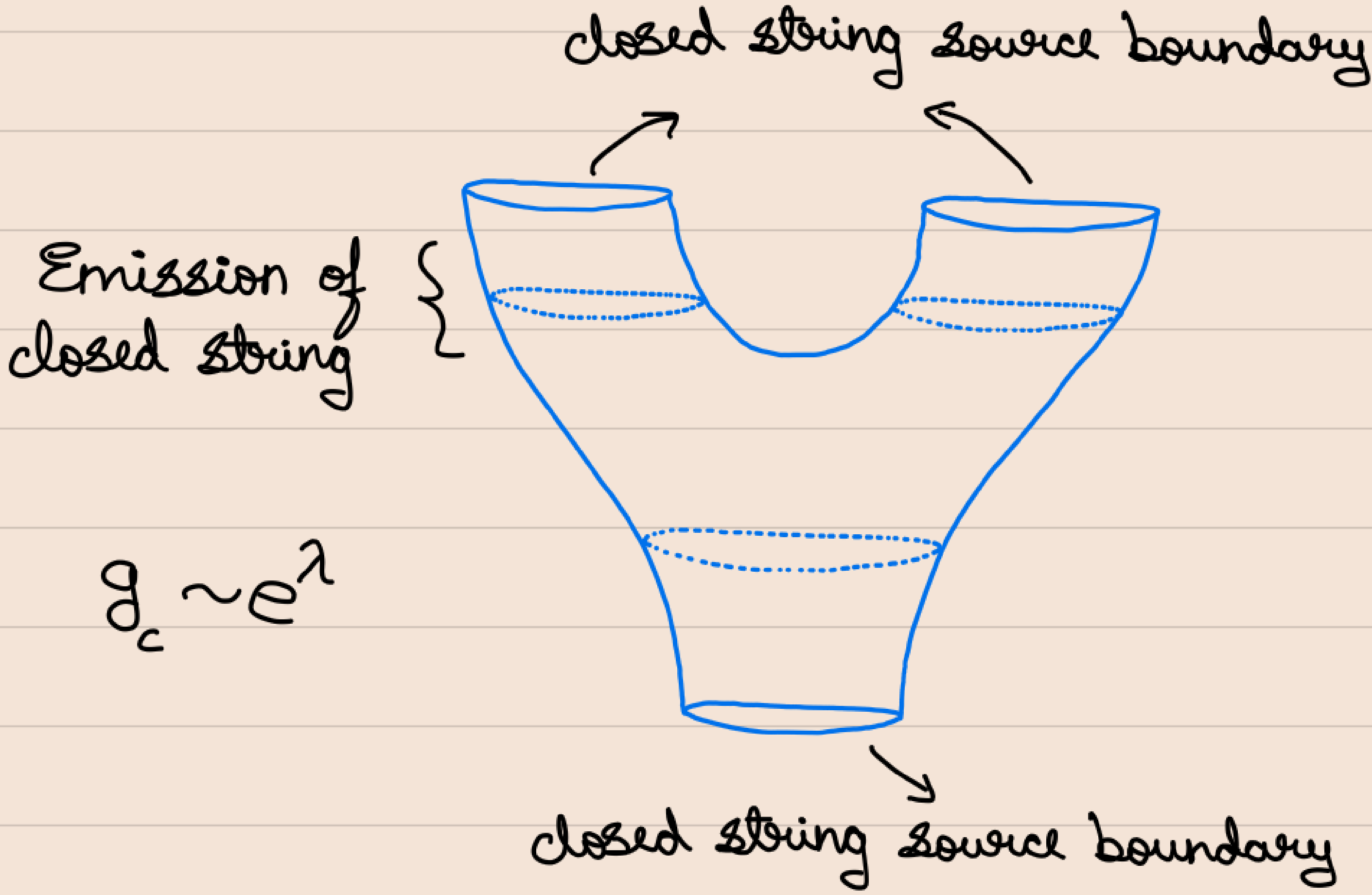


For closed strings



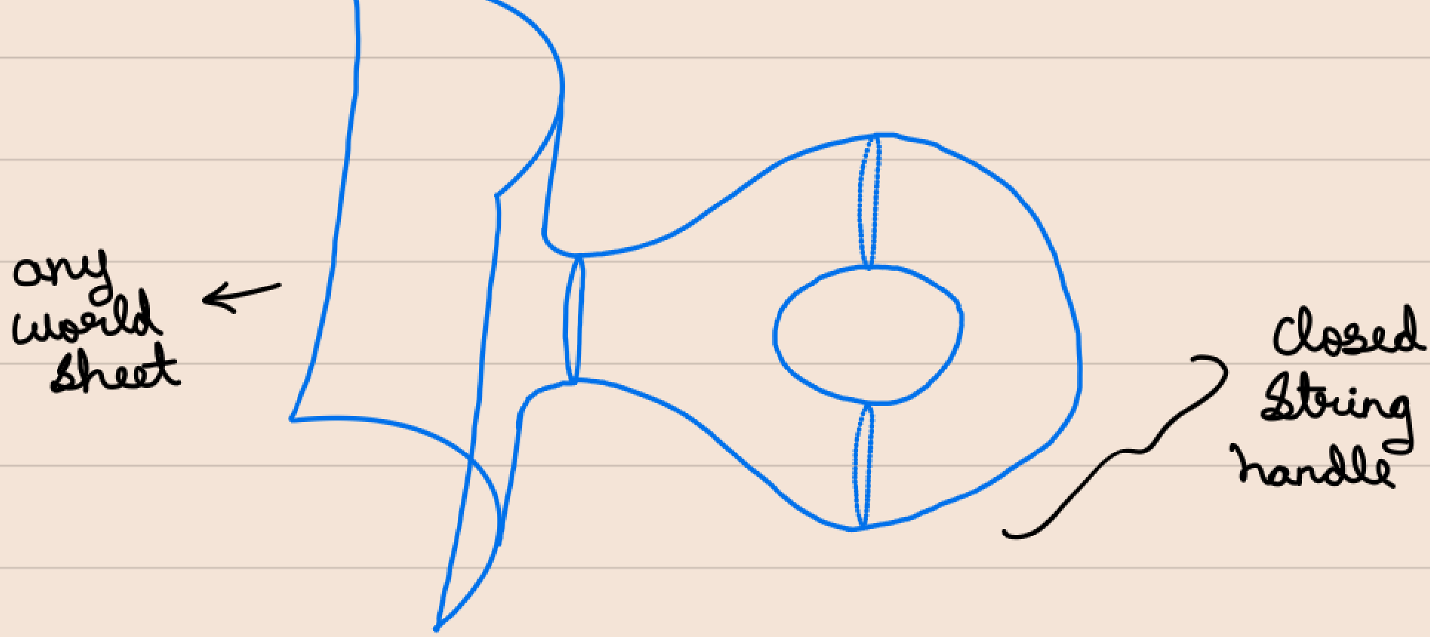
Adding a tube to a cylinder has a factor of $e^{2\lambda}$ to the path integral.

Thus process of closed string emission has amplitude proportional to e^{λ}.



$$g_c \sim e^{\lambda}$$

Adding a handle to any world sheet also has factor $e^{2\lambda}$ to it, as it amounts to emission & reabsorption of a closed string.



Counting of Couplings to String Sources

Source for closed strings are closed boundary loops & for open strings are boundary segments.

For convinience, we can choose the open string source boundaries to meet the string end-point boundary $\perp^r$ to each other.

Since boundary curvature diverges at the corners, so including corners in the Euler characteristic will be a problem. So, we can define

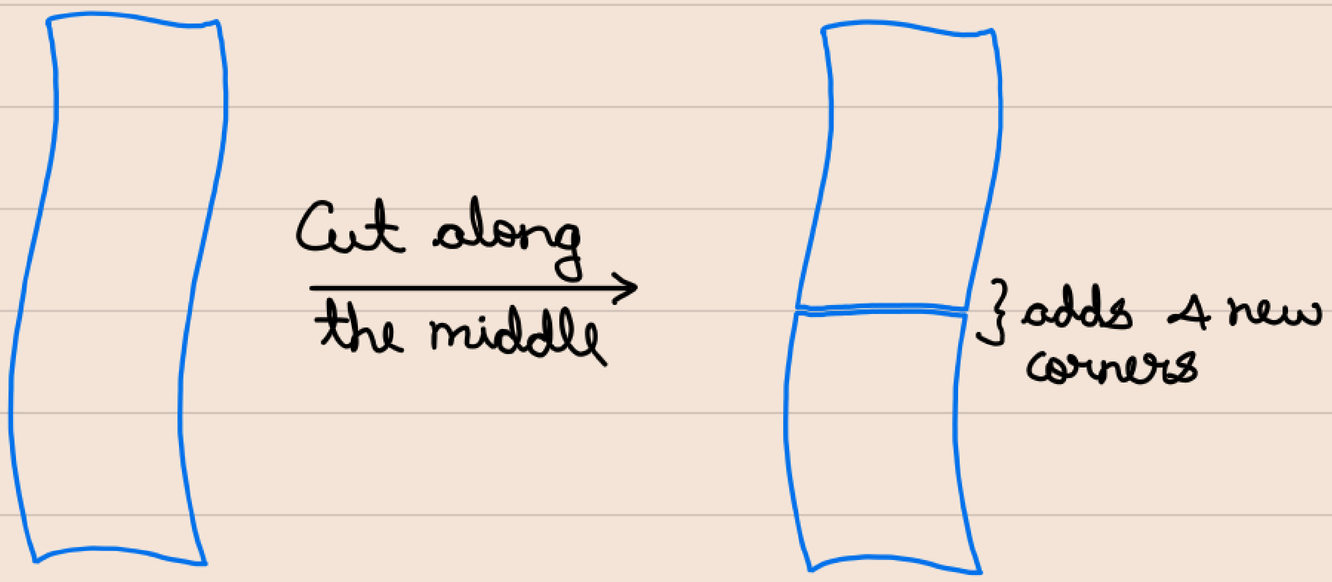
$$\chi = \tilde{\chi} + \frac{1}{4} n_{corner}$$

where '$\tilde{\chi}$' is the euler characteristic only over the smooth segments of boundary & n_{corner} is the number of corners created only by open string sources. Then, the appropriate weight for the path integral is

$$\text{“} e^{-\lambda\tilde{\chi}} = e^{-\lambda\chi} e^{\frac{\lambda}{4} n_{corner}} \text{”}$$

It is the appropriate weight since we need to maintain unitarity condition. It means that the 'λ' dependence must be unchanged if we cut through a world sheet tube/strip.

For open string, if we have a strip, with some χ & n_c initially.



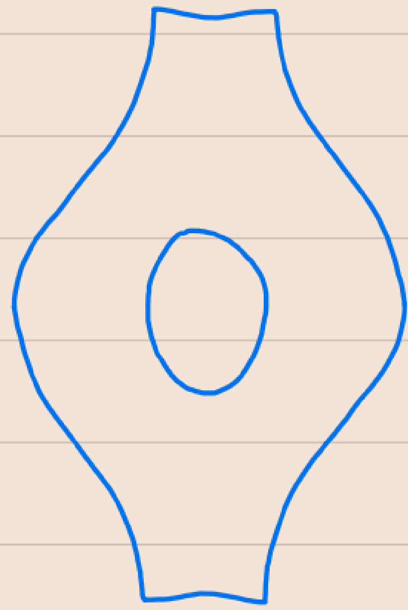
Now, $n_{corner} \longrightarrow n_{corner} + 4$, as two new string sources are added. Also $\tilde{\chi}$ remains invariant since the smooth part of the two sides of the cut have opposite orientation & hence will not contribute in '$\tilde{\chi}$'.

$\rightarrow \chi \longrightarrow \chi - 1$ on cutting

This is what we previously got when two physical sources were present (shown in the beginning). But in this case, we didn't add any physical source, it was just cutting a strip, a usual process we do to define a hilbert space formalism. Thus, $e^{-\lambda\tilde{\chi}}$ is a proper weight. χ & n_{corner} change in such a way that $\tilde{\chi}$ remains invariant.

But consider this process again.

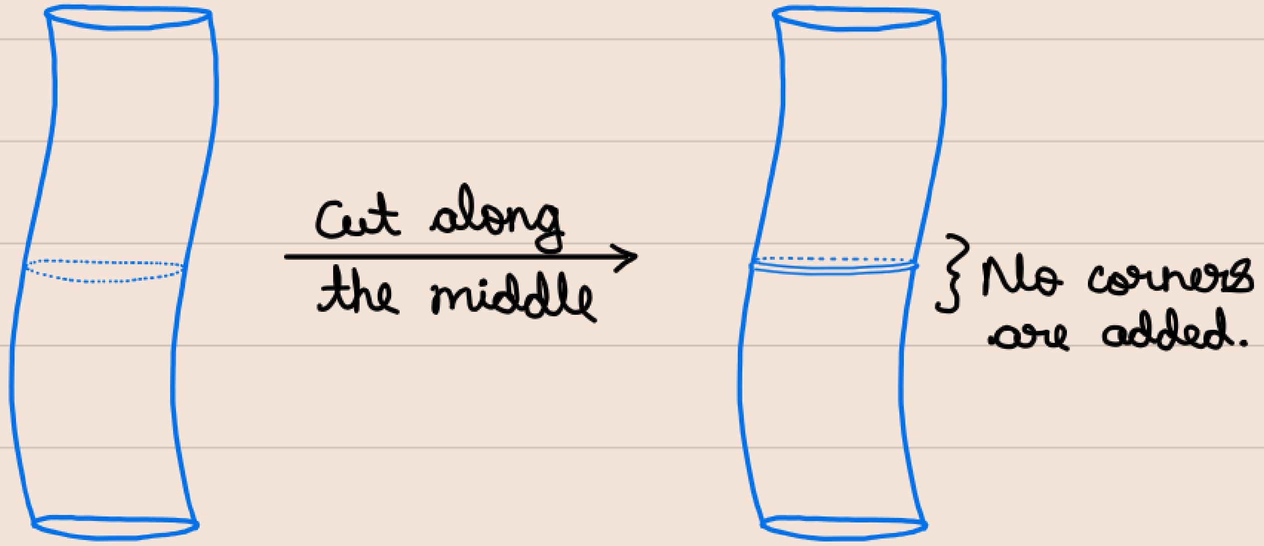
$$\xrightarrow[\text{by One}]{\text{`}\chi\text{' reduces}}$$



But n_{corner} doesn't change. So $\tilde{\chi}$ will reduce by one.

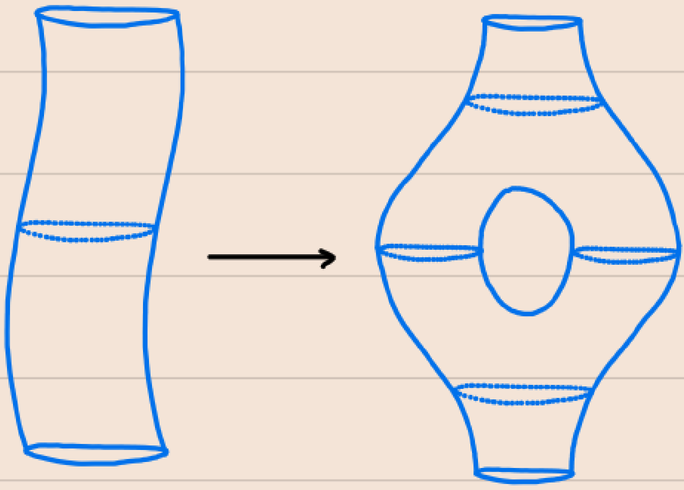
Thus this process will have usual amplitude proportional to 'e^{λ}' as before.

For closed string,



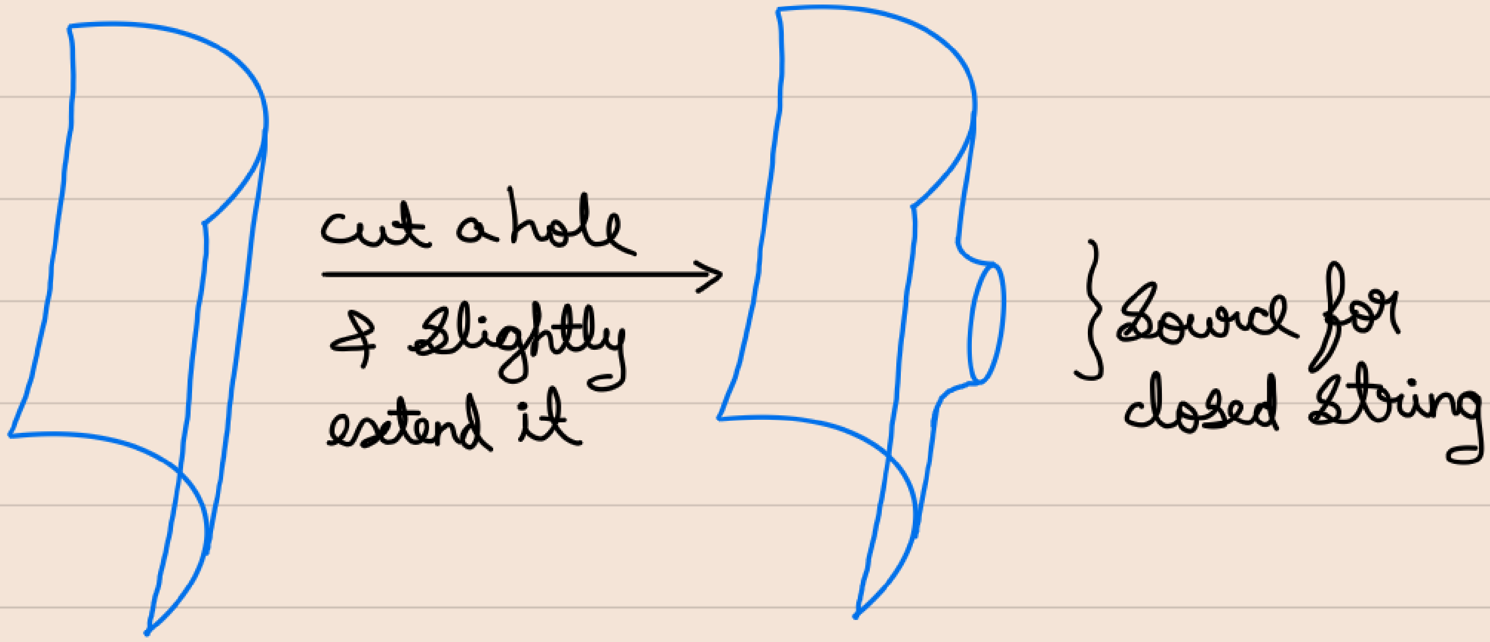
Now $\chi = \tilde{\chi}$. So here, $\tilde{\chi}$ remains invariant since the smooth part of the two sides of the cut have opposite orientation, Also same applies for 'χ' as this can be calculated fully due to no divergences in the corner term, so by the same arguement as that of $\tilde{\chi}$, even χ remains invariant.

In the below process, no n_c is added or removed, so $\chi = \tilde{\chi}$, hence we get the usual coupling constant for the weight $e^{-\lambda\tilde{\chi}}$.



Creation of new Sources

Consider cutting a hole in the world sheet & converting it as a source for the closed string,

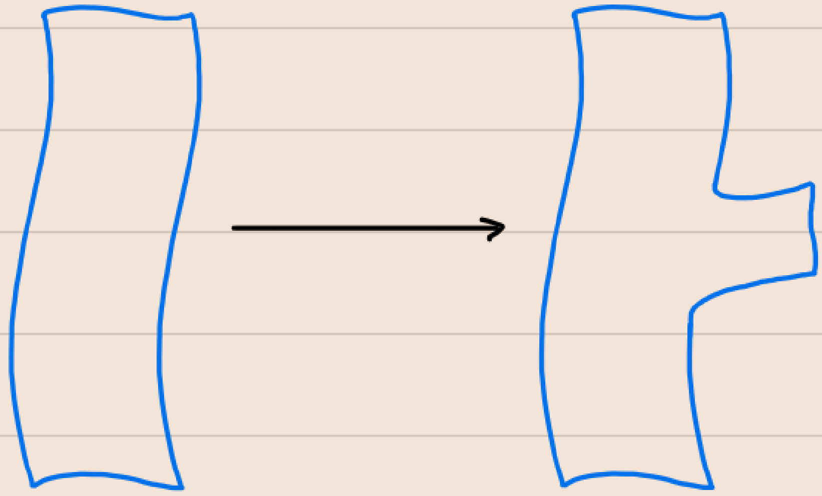


This changes $\chi \rightarrow \chi - 1$. But no source boundary corner is added, thus

$$e^{-\lambda\tilde{\chi}} \longrightarrow \underbrace{e^{\lambda}}_{g_c} e^{-\lambda\tilde{\chi}}$$

$\Longrightarrow$ we recover $g_c \sim e^{\lambda}$

Consider adding a new strip as a source for open string,



Here we need $\chi \to \chi$, $n_{corner} \to n_{corner} + 2$

$$e^{-\lambda\tilde{\chi}} = e^{-\lambda\chi} e^{\frac{\lambda}{4} n_{corner}} \longrightarrow e^{-\lambda\chi} e^{\lambda/2} e^{\frac{\lambda}{4} n_{corner}}$$

$$\longrightarrow e^{-\lambda\tilde{\chi}} \longrightarrow e^{\lambda/2} e^{-\lambda\tilde{\chi}}$$

Thus $\quad g_o \sim e^{\lambda/2}$.

So finally, we have established that

$$e^{-\lambda\tilde{\chi}} = e^{-\lambda\chi} e^{\frac{\lambda}{4} n_{corner}}$$

is the appropriate weight to maintain the unitarity condition.